

Hoofdstuk 3

Lesing 3.1

①

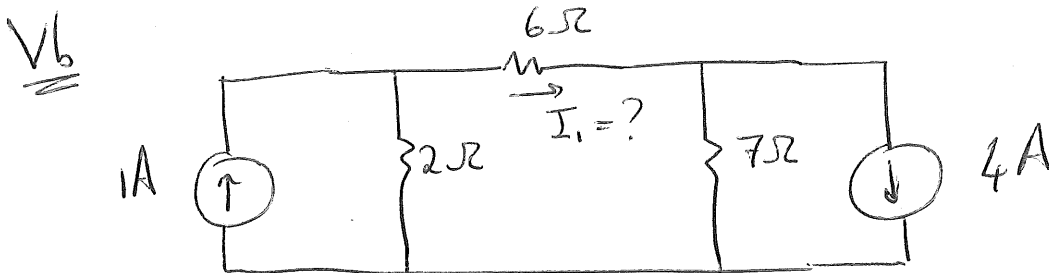
* Node - analyse.

→ Node analyse is gekoppel aan KCL.

→ Node analyse is som van strome, maar strome word geskryf in terme van spanning en weerstand.

$$\therefore \sum I_{IN} = \sum I_{UIT}$$

$$\rightarrow \sum_i \frac{V_i}{R_i} = \sum_j \frac{V_j}{R_j}$$



* Hoeveel nodusse? $\Rightarrow 3$.

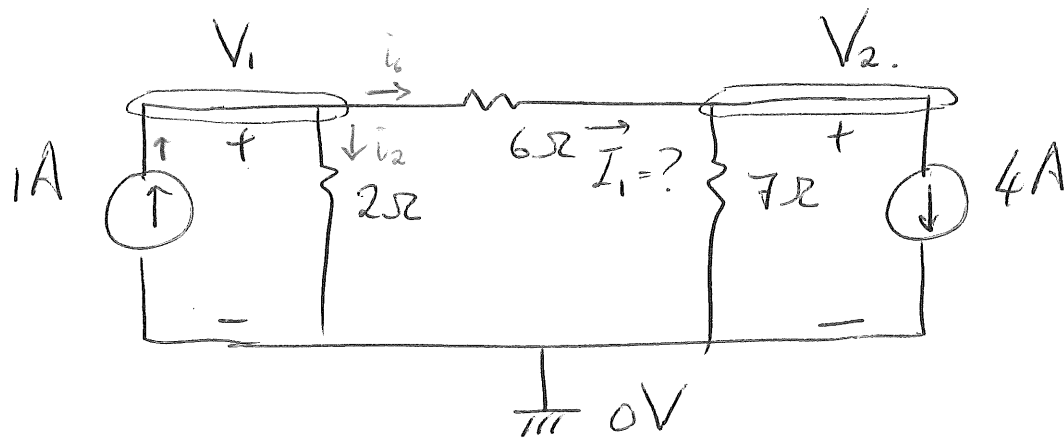
* Waar is verwysingspanning (0V)?

- Onderste node word verwysing.

- KCL word gedoen by ander 2 nodusse.

* Definies node-spanning by ander nodusse: V_1 en V_2 .

(2)



(Node-spanning is altyd $\oplus$ by node en $\ominus$ by 0V!)

① KCL by node ①.

$$1A = i_2 + i_6 \quad (\sum I_{IN} = \sum I_{OUT})$$

$$1A = \frac{+V_1}{2\Omega} + i_6$$

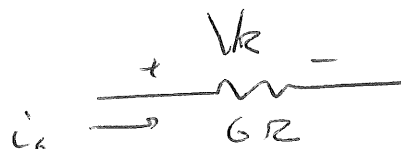
* V_1 is spanning oor 2Ω .

* i_2 vloei van $\oplus$ na $\ominus$ deur 2Ω .

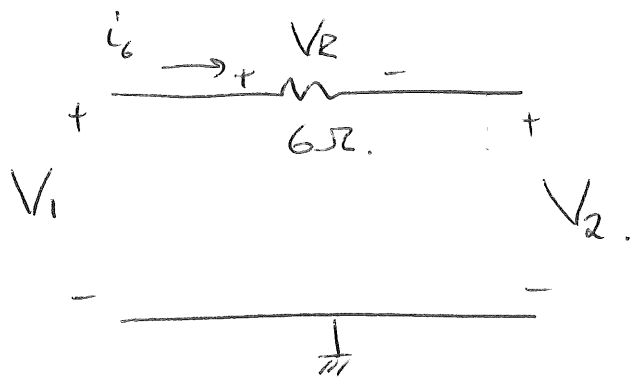
* Hoe kan ons i_6 skryf in V_1 en V_2 en 6Ω ?

$$* \text{ Ons weet } i_6 = \frac{+V_R}{6\Omega}$$

waar V_R die spanning oor die 6Ω is.



3



→ KVL: $-V_1 + V_R + V_2 = 0.$

∴ $V_R = V_1 - V_2.$

Dus: $i_6 = \frac{+V_R}{6\Omega} = \frac{V_1 - V_2}{6}.$

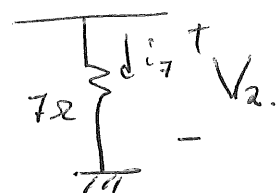
* KCL by node ① is dus

$$1 = \frac{V_1}{2} + \frac{V_1 - V_2}{6} \dots \dots \dots \underline{\underline{1}}$$

① KCL by node ②.

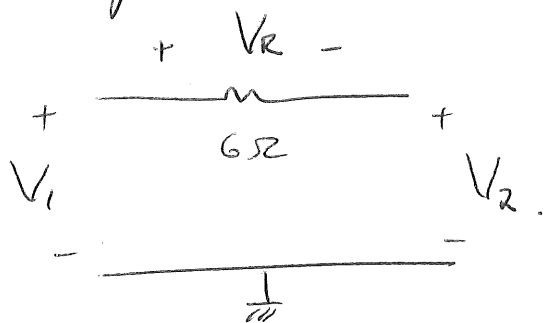
$i_6 = i_7 + 4$

$\frac{V_1 - V_2}{6} = \left(\frac{V_2}{7} \right) + 4 \dots \dots \dots \underline{\underline{2}}$



Met gelyktydige oplossing kan ons vergelykings
1 en 2 gebruik om V_1 en V_2
 op te los.

① KCL by node ②: Alternatief. ④



$$V_R = V_1 - V_2.$$

$$\begin{aligned} \vec{i} &= \frac{+V_R}{6\Omega} \\ &= \frac{V_1 - V_2}{6} \end{aligned}$$

$$\begin{aligned} \vec{i} &= \frac{-V_R}{6\Omega} \\ &= \frac{V_2 - V_1}{6} \end{aligned}$$

Dus, stroom in node-spannings kan geskryf word as:

$$i = \frac{V_{\text{FROM}} - V_{\text{TO}}}{R}$$

⇒ V_{FROM} is spanning waarvandaan die stroom vloei.

⇒ V_{TO} is spanning waarna stroom vloei.

⇒ $V_{\text{FROM}} - V_{\text{TO}}$ gee spanningsval oor weerstand in rigting waarin stroom vloei.

KCL by node (2)

$$0 = \frac{V_2 - V_1}{6\Omega} + \frac{V_2}{7\Omega} + 4.$$



Node - analyse.

1 → kies verwysing spanning (oV)

2 → allokies node-spannings
(⊕ by node ; ⊖ by oV)

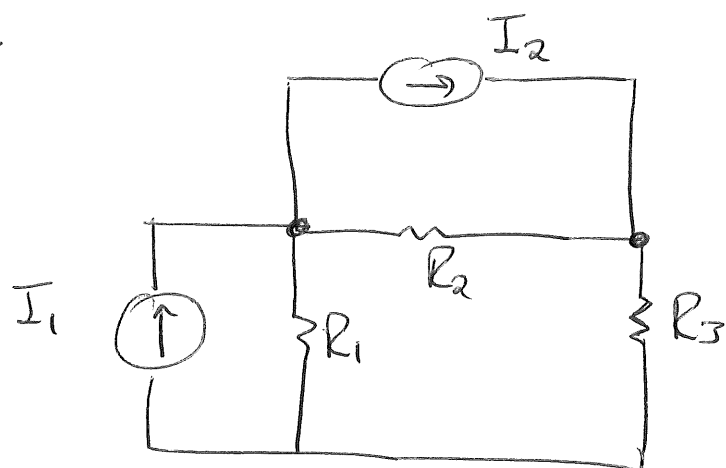
3 → KCL by elke node.

* Skryf strome its spannings en R'e, behalwe as stroom bekend is.

* Gewoonlik kies ons dat alle strome uit node vloei. $\therefore \sum I_{uit} = 0.$

V/L

6



V6

